

From Literature Retrieval to Meta-Analysis: A Reproducible AI-Assisted Framework for Evidence Synthesis

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Abstract

The integration of artificial intelligence (AI), particularly active learning and large language models (LLMs), into systematic reviews offers unprecedented efficiency gains. However, evaluating these tools in isolation obscures how algorithmic errors compound across the synthesis pipeline. This paper presents and empirically evaluates an end-to-end workflow for AI-assisted evidence synthesis, examining how errors introduced at the screening and extraction stages affect the final meta-analysis results. We propose a unified evaluation methodology that quantifies performance across four interconnected dimensions: retrieval consistency, screening efficiency (WSS@95), extraction agreement (κ), and the downstream impact of extraction errors on pooled effect sizes and heterogeneity (I^2). We demonstrate that while AI can dramatically reduce the screening burden, uncorrected LLM extraction errors systematically distort meta-analytic weighting, inflating heterogeneity and widening confidence intervals. The findings provide concrete guidance on when AI tools are safe to use autonomously and when human validation is mandatory, offering a practical framework for reviewers seeking to adopt AI assistance without compromising the reliability of their synthesized evidence.

1 Introduction

Systematic reviews remain one of the most time-intensive forms of research. Across the health and life sciences, the social sciences, the humanities, and engineering, researchers are expected to map rapidly evolving fields, evaluate interventions, and synthesize disparate findings into coherent, actionable conclusions. Systematic reviews and meta-analyses have long provided the methodological rigor required for these tasks, establishing themselves as the highest level of evidence in many disciplinary hierarchies. Yet, the traditional workflow—involving manual database searches, the screening of thousands of titles and abstracts by two

independent reviewers, and the painstaking extraction of data from full-text PDFs—is often characterized by massive time investments, significant human error, and a growing mismatch between the pace of publication and the pace of synthesis.

The methodological consequence of this mismatch is clear: if evidence synthesis is to remain feasible and rigorous in an era of information abundance, the integration of computational and AI-assisted tools must be treated as a core component of research design rather than an optional convenience. The scientific community has begun to recognize this imperative. Funding bodies, journals, and methodological organizations are increasingly acknowledging the need for AI-assisted approaches, while simultaneously calling for transparency, validation, and the preservation of human oversight in the review process [Fleming et al., 2025].

The emergence of machine learning and natural language processing has created unprecedented opportunities to automate the most labor-intensive stages of systematic reviewing. Active learning frameworks, such as ASReview, have demonstrated that human-in-the-loop algorithms can drastically reduce the number of abstracts that must be read to identify relevant studies, often achieving near-complete recall while requiring reviewers to screen only a fraction of the corpus [van de Schoot et al., 2021]. More recently, the advent of large language models (LLMs) like GPT-4 and Claude has opened the door to zero-shot abstract screening and automated data extraction from full texts, tasks that previously required extensive human labor [Mahmoudi et al., 2025, Kim et al., 2025]. However, the central challenge is not simply the availability of these AI tools, but the capacity to align them with meaningful research questions, ethically acceptable screening strategies, and defensible validation procedures. Crucially, any such workflow must proceed with explicit validation and human verification at each stage.

Furthermore, the data collection phase of evidence synthesis is undergoing a parallel transformation. The rise of open bibliographic metadata catalogs, such as OpenAlex and Crossref, alongside programmable APIs, allows researchers to bypass proprietary, siloed databases and construct comprehensive corpora programmatically and reproducibly [Delgado-Quirós and Ortega, 2024, Culbert et al., 2025]. This broader framing is especially relevant because an AI-assisted review is only as good as the corpus it operates on. The researcher does not encounter a neutral, perfect set of literature, but rather a dataset shaped by indexing practices, API limitations, and the precision of search query formulation. Recognizing this dependency is the first step toward rigorous AI-assisted synthesis.

In adjacent work on text analysis, Grimmer, Roberts, and Stewart emphasize that data construction is inseparable from theory, representation, and validation; there is no value-free corpus construction, and the suitability of a method depends on the task at hand [Grimmer et al., 2022]. Those insights are highly applicable to AI-assisted systematic reviews. The movement from manual to AI-assisted synthesis is not a simple upgrade; it is a methodological shift that introduces new forms of efficiency alongside new forms of risk. The researcher

who delegates screening to an active learning model or extraction to an LLM must understand the assumptions, failure modes, and validation requirements of those tools as clearly as a researcher who designs a survey instrument or chooses a statistical model.

This paper develops that argument in the context of AI-assisted systematic reviews and research synthesis across a wide range of academic disciplines. The aim is not to provide a comprehensive technical manual on specific LLM architectures or API wrappers. Nor is it to defend an unrestricted, fully automated approach to literature reviews that removes human judgment from the process. Instead, the goal is to provide a methodologically grounded account of how AI tools can be understood as distinct stages within a broader research workflow. That workflow begins with a substantive question and API-based programmatic literature collection, proceeds through active learning or LLM-based screening, and continues through automated data extraction, validation, preliminary synthesis, and ethical reporting. Understood in these terms, AI-assisted synthesis is less a magic bullet than a practical and epistemic bridge between overwhelming literature environments and rigorous, synthesized evidence.

Crucially, the methodological literature evaluating these tools has largely treated them as isolated components. Studies typically evaluate the recall of a screening algorithm or the accuracy of an extraction prompt in a vacuum. What is critically missing from the literature is a practical account of how errors at one stage of the pipeline—such as false positives in automated screening or incorrect sample sizes extracted by an LLM—affect the final meta-analysis results. This paper addresses that gap by presenting and empirically evaluating an integrated workflow for AI-assisted evidence synthesis, and by providing concrete recommendations for reviewers on when AI assistance is reliable and when human validation is essential.

The remainder of the paper proceeds as follows. Section 2 reviews the literature on active learning, LLM-based extraction, open bibliographic APIs, and the methodological status of automated reviews. Section 3 outlines the conceptual architecture of the proposed framework for AI-assisted evidence synthesis. Section 4 examines core design principles: query formulation, programmatic literature collection, and corpus construction. Section 5 details the mechanics and theory of screening models, including active learning and LLM classification. Section 6 addresses structured information extraction, validation, data quality, hallucination risks, and ethical reporting standards. Section 7 details the statistical and methodological principles of synthesis and meta-analysis, alongside a discussion of technical failure modes and structural biases specific to AI-assisted workflows. Section 8 details the experimental design used to evaluate the framework. Section 9 presents the empirical evaluation across four dimensions (retrieval, screening, extraction, synthesis). Section 10 discusses the implications of these findings for end-to-end error propagation and pipeline validation. Section 11 concludes.

2 Literature Review: AI and Automation in Evidence Synthesis

The methodological literature relevant to AI-assisted systematic reviews spans several adjacent fields rather than a single unified tradition. Understanding these literatures in relation to each other is essential for situating any specific tool or technique within a broader methodological context.

2.1 Machine Learning for Screening

In the domain of machine learning for evidence synthesis, the focus has historically been on the abstract screening phase. Van de Schoot and colleagues introduced ASReview, an open-source platform that implements active learning to accelerate the screening of large textual datasets [van de Schoot et al., 2021]. Their simulation studies demonstrated that active learning can yield far more efficient reviewing than manual screening while maintaining high recall, often identifying 95% of relevant studies after screening only 5–10% of the corpus. This work established a foundational principle: computational tools should augment rather than replace human judgment. The human reviewer remains in the loop, iteratively labeling a small set of papers to guide a prioritization model that continuously improves its predictions. This principle is central to rigorous review methodology and distinguishes responsible AI-assisted reviewing from fully automated, unsupervised approaches.

Building on this foundation, Teijema and colleagues examined the performance of active learning using deep learning-based models compared to traditional machine learning approaches, finding that switching between models during the review process can improve efficiency [Teijema et al., 2023]. This line of research underscores that no single algorithmic approach is universally optimal; the choice of model should be guided by the characteristics of the corpus and the nature of the research question.

2.2 Large Language Models in Systematic Reviews

A second, rapidly expanding body of work concerns the application of Large Language Models (LLMs) to systematic reviews. The capabilities of models like GPT-4 and Claude have prompted researchers to explore their utility across multiple stages of the review process, from formulating search queries to screening abstracts and extracting data from full texts. Recently, Qiao et al. evaluated the use of LLMs for zero-shot abstract screening, demonstrating that while these models can achieve high recall without initial training data, they frequently suffer from low precision, generating false positives that require subsequent human filtering [Qiao et al., 2024].

Mahmoudi et al. critically assessed LLM performance in data extraction for systematic

reviews, noting both the potential for high accuracy on structured tasks and the persistent risks of nuanced errors and hallucination on complex, contextually sensitive information [Mahmoudi et al., 2025]. Similarly, Kim et al. explored customized GPT-4 models for extracting study data and assessing risk of bias in systematic reviews, highlighting that while AI can achieve impressive accuracy on well-defined extraction tasks, human oversight remains indispensable for ensuring the integrity of the final dataset [Kim et al., 2025]. Lieberum and colleagues conducted a scoping review of LLM applications in systematic reviews, concluding that while the technology is rapidly maturing, it is not yet ready for unsupervised use and requires robust validation frameworks [Lieberum et al., 2025].

These contributions are vital because they resist the idea that LLMs are infallible. Rather, they treat LLM outputs as probabilistic representations that require validation against the source material. The risk of hallucination—the generation of plausible but factually incorrect information—is particularly acute in the context of data extraction, where an LLM might invent a sample size or misattribute a finding if the prompt is insufficiently constrained.

2.3 The Gap: End-to-End Error Propagation

While the aforementioned literatures have rigorously evaluated individual components of the AI-assisted review process—such as the recall of active learning models or the categorical accuracy of LLM extraction—they have largely treated these tools as isolated interventions. What remains critically under-theorized is the concept of *end-to-end error propagation*. If an automated deduplication script falsely merges two distinct studies, or if an LLM hallucinates a subgroup sample size as the total study population, how do these errors interact and compound? Crucially, how do uncorrected numeric extraction errors systematically distort the pooled effect sizes and heterogeneity statistics of a meta-analysis? This paper addresses this gap by presenting a unified evaluation methodology that traces error propagation from initial retrieval through to the final forest plot.

2.4 Open Bibliographic APIs and the Infrastructure of Retrieval

A third line of scholarship deals with the infrastructure of literature retrieval. The shift toward open science has elevated platforms like OpenAlex, Crossref, Semantic Scholar, and Europe PMC as primary sources for bibliographic metadata. These platforms offer free, programmatic access to millions of scholarly records via well-documented APIs, representing a significant democratization of literature access [Priem et al., 2022].

Delgado-Quirós and Ortega analyzed the completeness of metadata across eight free-access scholarly databases, noting the viability of OpenAlex and Crossref as alternatives to proprietary systems like Web of Science or Scopus, while also identifying gaps in abstract coverage for some disciplines [Delgado-Quirós and Ortega, 2024]. Culbert and colleagues

further analyzed OpenAlex’s reference coverage compared to Web of Science and Scopus, underscoring its utility for programmatic, API-driven corpus construction [Culbert et al., 2025]. This literature is important because it situates AI screening within the broader context of open data access: an AI model can only synthesize the literature it is fed, making the quality and comprehensiveness of the API retrieval process paramount.

2.5 Reporting Standards and Reproducibility

A fourth literature addresses the methodological standards and reporting guidelines governing systematic reviews. The PRISMA 2020 statement provides the definitive framework for reporting systematic reviews, emphasizing transparency in search strategies, study selection, data extraction, and synthesis [Page et al., 2021]. The PRISMA-S extension specifically addresses the rigorous reporting of literature searches, which is critical when using programmatic API retrieval [Rethlefsen et al., 2021]. For environmental and broader life sciences, frameworks such as ROSES (RepOrting standards for Systematic Evidence Syntheses) serve a similar function [Pullin et al., 2018]. As AI tools are integrated into these workflows, adhering to these guidelines becomes critical for ensuring reproducibility and scientific trust. Hardwicke and colleagues’ empirical assessment of transparency and reproducibility practices further underscores the need for open code and data sharing, including the sharing of AI prompts and model configurations used in systematic reviews [Hardwicke et al., 2020].

2.6 Ethics and Epistemic Risk

Finally, there is a growing literature on the ethical and epistemic risks of generative AI in research contexts [Mittelstadt, 2019, Harris et al., 2023]. Issues of algorithmic bias, hallucination, and data privacy are central concerns [Bender et al., 2021]. As researchers delegate screening and extraction tasks to LLMs, they must navigate the tension between efficiency and the risk of introducing systematic errors into the synthesized evidence base. The broader internet research ethics literature reinforces the importance of contextual decision-making rather than one-size-fits-all prescriptions [Markham et al., 2012], a principle that applies equally to the use of AI in evidence synthesis.

3 A Framework for AI-Assisted Evidence Synthesis

We propose a unified, conceptual architecture for AI-assisted evidence synthesis that moves sequentially from query formulation to statistical meta-analysis. The framework is built on the principle that computational tools should augment rather than replace human judgment, requiring explicit validation at each stage.

The architecture consists of four primary stages: 1. **Literature Retrieval and Corpus Construction:** Utilizing open APIs to programmatically retrieve, deduplicate, and visualize a comprehensive corpus based on structured Boolean queries. 2. **Screening Strategies:** Applying active learning or LLM-based zero-shot classification to efficiently identify relevant studies from the retrieved corpus, minimizing the human screening burden while maximizing recall. 3. **Structured Information Extraction:** Deploying LLMs with strict schema definitions to extract quantitative and qualitative variables from full-text PDFs, followed by human validation of the extracted data. 4. **Statistical Synthesis and Meta-Analysis:** Computing pooled effect sizes, assessing heterogeneity, and evaluating publication bias using the extracted, validated dataset.

By integrating these stages into a single workflow, the framework ensures that the provenance of every data point—from the initial API query to the final forest plot—is fully traceable and reproducible.

3.1 The Cross-Disciplinary Imperative

The practical appeal of AI-assisted systematic reviews derives from the universal challenge of information overload across a wide range of academic disciplines. While the methodology of the systematic review originated in the health sciences, the need to rigorously synthesize literature is now ubiquitous.

In the health and life sciences, researchers must aggregate findings from hundreds of randomized controlled trials, epidemiological studies, or genomic analyses to inform clinical guidelines or public health policy. The sheer volume of daily publications makes manual screening a significant barrier to timely evidence synthesis [Wang et al., 2025]. AI tools, particularly active learning for screening and LLMs for extracting specific trial characteristics (e.g., sample size, dosage, primary outcomes, risk of bias assessments), offer a direct solution to this bottleneck [Guo et al., 2024]. The PRISMA framework provides a well-established reporting standard for these reviews, and AI tools must be deployed in a manner consistent with its requirements.

In the social sciences, evidence synthesis often takes the form of scoping reviews, qualitative meta-syntheses, or systematic reviews of policy interventions, educational outcomes, or psychological phenomena. Here, the literature is frequently characterized by diverse methodologies, varying terminologies, and complex theoretical frameworks that resist simple keyword-based retrieval. AI tools can assist in harmonizing these diverse sources, using LLMs to categorize qualitative findings or extract nuanced contextual variables—such as the theoretical framework employed, the country of study, or the specific population targeted—that traditional keyword searches might miss [Fütterer et al., 2026, Garzón et al., 2025].

In the humanities, researchers conduct literature reviews to map the development of theoretical concepts, trace the intellectual genealogy of a scholarly debate, or synthesize

interpretive analyses of cultural artifacts. While API-based retrieval and PDF extraction may be less complete for some humanities sub-fields due to the prevalence of monographs and older archival materials, the fundamental need to systematically search, screen, and extract information from a large corpus remains. Open APIs allow for comprehensive mapping of the scholarly literature on a given topic across decades, while LLMs can assist in thematic extraction and narrative synthesis.

In engineering and the applied sciences, researchers conduct systematic reviews to map the state of the art in a specific technology, evaluate the performance of competing methods, or identify gaps in the literature that warrant further investigation. The literature in these fields is often highly technical and quantitative, making structured data extraction a natural application for LLM-based tools, including fully automated end-to-end review systems [Al-Marridi et al., 2025].

What these diverse applications share is not a common substantive domain, but a common workflow of evidence construction. In each case, the researcher begins with a corpus of literature and asks whether AI tools can accurately and efficiently identify relevant studies and extract meaningful data. This commonality is what makes AI-assisted synthesis teachable across a wide range of disciplines.

4 Literature Retrieval and Corpus Construction

4.1 From Research Question to Search Strategy

A robust AI-assisted review begins with a research question rather than with a machine learning model. The literature on systematic reviews consistently warns against source-led or tool-led design. Question-led design determines the search strategy, the selection of bibliographic APIs, and the formulation of Boolean queries. Structured frameworks help translate a research question into a Boolean query. In health sciences, the PICO framework (Population, Intervention, Comparison, Outcome) provides a structured approach to translating a clinical question into a search strategy. In the social sciences, analogous frameworks such as SPIDER (Sample, Phenomenon of Interest, Design, Evaluation, Research type) serve a similar function. The choice of framework should reflect the nature of the research question and the type of evidence being sought.

A Boolean query translates a conceptual question into a string the API can search. The operators **AND** (narrows results), **OR** (broadens results), and ***** (wildcard for word variants) are foundational. A common mistake is designing queries that are either too broad (returning hundreds of thousands of unrelated papers) or too narrow (returning zero or very few papers). The heuristic “Goldilocks Rule” suggests aiming for 200 to 2,000 results. The keyword network can often indicate whether the query is on target.

4.2 Querying Scholarly APIs

Open APIs such as OpenAlex, Crossref, and Europe PMC offer programmatic access to millions of scholarly records. Each API has distinct strengths. OpenAlex covers all disciplines with approximately 250 million works and around 87% abstract coverage, making it a strong starting point. Crossref provides robust publisher metadata, while Europe PMC excels in the life sciences and biomedicine with near-complete abstracts and full-text search capabilities for open-access articles.

Querying these APIs requires careful design. Researchers must translate their conceptual questions into precise search strings, utilizing appropriate filters (e.g., publication year, document type, open access status) and handling pagination to retrieve comprehensive results. APIs never return all results at once; they split them into pages. Two common pagination styles are offset pagination (used by Crossref and Europe PMC) and cursor pagination (used by OpenAlex, which is generally faster and more reliable for large corpora).

APIs also enforce rate limits to prevent server overload. Exceeding the limit results in a “429 Too Many Requests” status code. Best practices for being a polite API user include adding short sleep intervals between requests, using the polite pool (e.g., adding an email address to OpenAlex and Crossref requests), requesting the maximum allowed records per page, and caching results locally to avoid re-downloading data. While OpenAlex, Crossref, and Europe PMC do not require authentication keys for basic use, other APIs like Scopus or Web of Science require institutional licenses and keys, which must be handled securely.

4.3 Metadata and Deduplication

When OpenAlex returns a record, key fields include the title, publication year, DOI, authorships, concepts (AI-assigned topic tags), citation counts, open-access URLs, and the abstract. Notably, OpenAlex stores abstracts as an inverted index—a dictionary mapping each word to its position—rather than plain text. This reduces storage and speeds up search, but requires reconstruction into readable text for downstream analysis.

A further advantage of API-based retrieval is the ability to programmatically combine results from multiple sources and apply automated deduplication. When querying multiple APIs or running multiple queries, the same paper often appears more than once due to indexing overlap, preprint versus published versions, or slight DOI variations. Deduplication algorithms that match on DOIs, titles (normalized by removing punctuation and converting to lowercase), or a combination of metadata fields can automatically identify and merge these duplicate records, producing a clean, unduplicated corpus ready for screening.

4.4 Visualizing the Corpus

Before screening, visualizing the retrieved corpus can provide valuable insights. A keyword co-occurrence network, where nodes represent concept tags and edges represent their co-occurrence in papers, can reveal the thematic structure of the literature. Central nodes indicate core concepts, peripheral nodes suggest niche topics, and bridge nodes highlight interdisciplinary connections. Warning signs in the network, such as a single giant cluster without sub-structure or dozens of isolated nodes, may indicate that the query is too broad or too narrow, respectively.

Additionally, plotting the publication year distribution can reveal the trajectory of the field. Exponential growth suggests a rapidly emerging field, while a flat or declining trend indicates maturity. Sharp spikes may point to landmark papers or crisis events. If older literature is missing, it may be necessary to include historical synonyms in the query.

4.5 Reproducibility and Export

The advantage of programmatic literature collection over manual database searching is reproducibility: an API query can be saved, shared, and rerun exactly. However, because APIs evolve and results may change over time, reproducibility requires logging queries and timestamps—a practice that robust workflows must support automatically. The PRISMA-S extension requires that the API used, the exact Boolean query string, all applied filters, the date of the query, and the total number of records before and after deduplication be reported [Rethlefsen et al., 2021]. Exporting the corpus in appropriate formats, such as CSV for data cleaning or RIS for reference managers like Zotero, facilitates the transition to the screening phase.

5 Screening Strategies

5.1 Active Learning and Text Vectorization

Once a deduplicated corpus is constructed, the researcher must configure the screening approach. Active Learning, as implemented in tools like ASReview, requires the researcher to iteratively label a small set of papers to guide a prioritization model [van de Schoot et al., 2021]. The algorithm trains a model to predict the relevance of the remaining corpus, constantly updating as the researcher provides more labels.

Active learning requires converting text into a numerical matrix, typically using Term Frequency–Inverse Document Frequency (TF-IDF). The weight of a term in a document is proportional to its frequency in the document and inversely proportional to its frequency across the corpus. This down-weights common words and highlights rare, domain-specific

terms, allowing the classifier to locate papers in a high-dimensional vector space [Salton and McGill, 1983].

Active learning relies on query strategies to determine which instances to present to the human for labeling. Common strategies include uncertainty sampling (querying instances the model is least certain about), query-by-committee (querying instances where an ensemble of models disagrees), and expected model change. In systematic reviews, uncertainty sampling is often used to rapidly surface the most likely relevant papers.

A key challenge in active learning is determining when to stop screening. The standard performance metric is Work Saved over Sampling (WSS@95), which measures the proportion of the corpus that does not need to be screened to achieve 95% recall compared to random screening [Cormack and Grossman, 2016]. In practice, reviewers often adopt stopping rules based on a consecutive sequence of irrelevant papers, balancing efficiency with completeness.

5.2 LLM-Based Screening and Prompting Paradigms

An alternative to active learning is LLM-based screening. There are three primary paradigms for applying LLMs to classification: zero-shot (applying criteria from the prompt alone with no labeled examples), few-shot (including a small number of labeled examples in the prompt for in-context learning), and fine-tuning (updating the model’s internal weights on a domain-specific labeled dataset). Zero-shot requires no training data but relies entirely on prompt clarity, while fine-tuning offers high performance on specialized text but requires significant resources.

LLM-based screening evaluates each abstract against explicit inclusion and exclusion criteria, providing a binary decision. While this approach can be faster and requires no initial labeling effort, it demands rigorous prompt engineering. Empirical evidence shows that LLM outputs are highly sensitive to superficial prompt variations, meaning different formulations can yield systematically different decisions [Mizrahi et al., 2024]. To ensure reproducibility, prompts must be pre-registered and reported verbatim in the methods section. Furthermore, LLM outputs are non-deterministic, necessitating fixed temperature settings (e.g., $T = 0$) for consistency.

6 Structured Information Extraction

6.1 Extraction Schema Design

The most complex phase of the AI-assisted workflow is data extraction from full-text PDFs. The researcher must first define an explicit, discipline-specific extraction schema—such as a PICO template for health sciences or a Thematic Synthesis template for the social sciences and humanities.

Using LLMs for extraction requires translating this schema into precise, structured prompts. The LLM must be instructed not only on what to extract, but on the required output format (e.g., JSON or a structured table) and the constraints on inference. This is a measurement decision. The prompt defines what counts as an observation and what form of comparability the resulting dataset will support. Recognizing this prompting layer as a site of potential measurement error is essential for responsible interpretation.

A well-formed screening justification can serve as a bridge to extraction. By articulating which PICO elements were present in the abstract during the screening phase, the LLM performs a form of proto-extraction. This screening-informed extraction can pre-populate templates, reducing cognitive load and improving consistency during synthesis.

6.2 Validation, Quality, and Ethical Reporting

6.3 The Necessity of Validation

Validation is the cornerstone of AI-assisted evidence synthesis. The fact that an LLM can rapidly populate a spreadsheet does not, by itself, show that the data is accurate or that the screening decisions are correct. AI-assisted extraction is probabilistic and requires auditing. Validation in this context has at least five distinct dimensions.

The first is screening validation: confirming that the active learning model or LLM has not erroneously excluded relevant studies. The second is extraction validation: checking that the LLM has accurately retrieved the requested variables from the full texts without hallucination. This requires the researcher to manually audit a representative subset of the extracted data against the original PDFs. The third is semantic validation: confirming that the extracted values are semantically consistent across studies. The fourth is coverage validation: assessing whether the API retrieval strategy captured a sufficiently comprehensive set of relevant literature. The fifth is analytic validation: determining whether the synthesized data behaves plausibly in summaries, distributions, or meta-analyses.

6.4 Measuring Agreement: Cohen’s Kappa

To validate AI screening, inter-rater reliability must be measured. The standard metric is Cohen’s Kappa (κ), which accounts for agreement expected by chance [Cohen, 1960]. Interpretation guidelines suggest that a κ below 0.60 generally requires criteria revision [Landis and Koch, 1977].

However, Kappa is highly sensitive to the prevalence of the positive class, a phenomenon known as the prevalence paradox [Feinstein and Cicchetti, 1990, Cicchetti and Feinstein, 1990]. When the proportion of included studies is very low, Kappa can be misleadingly low even when absolute percentage agreement is high. Best practice dictates reporting Kappa alongside positive agreement and negative agreement statistics to provide a transparent view

of divergence. Validation protocols should involve a human reviewer independently applying criteria to a stratified random sample, blinded to the AI’s decisions, to compute these metrics.

6.5 Mitigating Hallucination and Confabulation

When using LLMs for screening and extraction, researchers must distinguish between hallucination (generating plausible but factually incorrect content) and confabulation (generating internally consistent but externally false reasoning). Mitigating these risks requires strict prompt engineering, the use of lower temperature settings, and demanding a structured justification in the prompt. Forcing the model to articulate its reasoning allows the human reviewer to spot-check for these failure modes.

6.6 Ethical Reporting and PRISMA 2020

The integration of AI into systematic reviews necessitates transparent reporting. The PRISMA 2020 statement provides a comprehensive checklist for reporting the methods and findings of systematic reviews [Page et al., 2021]. For API-based reviews, the PRISMA 2020 flow diagram requires reporting records identified through databases separately from other methods, duplicates removed, records excluded at the abstract stage, and specific reasons for exclusion at the full-text stage.

As AI tools enter the workflow, reporting standards are adapting. Emerging consensus mandates the disclosure of the tool name, model architecture, exact prompt text, validation procedures, inter-rater reliability metrics, and the human oversight protocol [Fleming et al., 2025]. The distinction between “AI-assisted” (human verified) and “AI-conducted” (fully automated) screening is critical for credibility. The output of the screening phase is a methodological record with its own provenance, where every included record must be traceable back to specific API queries, deduplication rules, and screening decisions.

6.7 Technical Failure Modes and Structural Biases

The methodological discussion of AI-assisted systematic reviews often emphasizes research design, validation, and ethical considerations. However, in practice, a substantial share of error arises from technical failure modes and structural biases that affect the integrity of the resulting dataset.

A first category of failure concerns the instability of API endpoints and data schemas. Bibliographic APIs are not static; their data models, query syntax, and rate limits can change over time. A workflow that functions correctly today may produce different results or fail entirely after an API update.

A second category concerns the incomplete coverage and structural biases of open APIs. API-based retrieval inherits the biases of the published literature, including the file-drawer

problem, where studies with null or negative results are less likely to be published [Rosenthal, 1979]. This publication bias means the retrieved corpus may skew toward positive findings, often necessitating supplementary searches of grey literature. Furthermore, APIs exhibit language bias (over-representing English-language studies), geographic bias (over-representing research from high-income countries), and indexing bias (skewing toward established journals). These biases constrain generalizability and must be explicitly acknowledged.

A third category concerns the quality and consistency of LLM outputs. Even with carefully designed prompts, LLM outputs are non-deterministic and may vary across runs, requiring consistency checks. The same paper may yield different extracted values on different runs of the same prompt, introducing a form of non-deterministic measurement error. Researchers should run extraction multiple times on a sample of papers and assess the consistency of the outputs before applying the workflow to the full corpus.

A fourth category concerns PDF parsing failures. AI-assisted extraction may fail due to PDF structure variability, OCR noise, multi-column layouts, or ambiguous phrasing; therefore all outputs must be treated as provisional and subject to manual verification. When the PDF parser fails to extract clean text, the LLM receives garbled or incomplete input, leading to unreliable extraction outputs or hallucinated extractions based on fragmented sentences.

7 Statistical Synthesis and Meta-Analysis

7.1 Statistical Models of Synthesis and the Impact of Propagated Error

The final phase of the systematic review workflow is statistical synthesis. In a meta-analysis, the choice between a fixed-effect model (which assumes a single true effect size) and a random-effects model (which assumes a distribution of true effect sizes) dictates how study weights are calculated. In a random-effects model using the DerSimonian–Laird estimator, the weight of each study is determined by both its within-study variance (heavily dependent on the extracted sample size) and the between-study variance (τ^2) [DerSimonian and Laird, 1986].

When extraction errors propagate into the synthesis phase, they systematically alter these fundamental statistics. An LLM error that underestimates the sample size of a highly precise study will artificially reduce that study’s weight in the pooled estimate. Furthermore, random extraction errors across multiple studies will artificially inflate the observed heterogeneity (I^2 and Q statistics), leading the researcher to conclude that the literature is more inconsistent than it actually is [Higgins and Thompson, 2002].

By comparing the pooled effect sizes and heterogeneity statistics derived from an uncorrected AI-extracted dataset against those derived from a human-validated gold standard, researchers can quantify the exact magnitude of error propagation. This comparative syn-

thesis serves as the ultimate validation metric for the AI-assisted framework, demonstrating the precise boundaries of algorithmic reliability in evidence synthesis.

7.2 Disciplinary Applications

A key advantage of AI-assisted evidence synthesis is that it can support substantively diverse projects while relying on a common methodological logic. The workflow described in this paper is not tied to any single discipline; it is a general-purpose research infrastructure that can be adapted to the specific needs of any field.

In the health and life sciences, AI-assisted reviews are most immediately applicable to the synthesis of clinical trial evidence. Researchers can use open APIs to retrieve trial registrations and published results, iteratively label a small set of papers to guide a prioritization model for screening, and use LLMs to extract structured data on interventions, comparators, outcomes, and risk of bias using a PICO template.

In the social sciences, AI-assisted synthesis is particularly valuable for scoping reviews and systematic reviews of policy interventions. Researchers can use programmatic literature collection to retrieve literature on a specific policy domain, apply LLM-based screening to identify studies that evaluate the effectiveness of a particular intervention, and use a Thematic Synthesis template to pull data on study design, population, context, and outcomes.

In the humanities, AI-assisted synthesis can support intellectual history and bibliometric analysis. Researchers can use open APIs to construct comprehensive corpora of publications on a specific theoretical concept or historical period, apply LLMs to extract thematic and argumentative content, and use tools like bibliometrix to map the evolution of scholarly debates over time [Aria and Cuccurullo, 2017].

In engineering and the applied sciences, AI-assisted synthesis is well-suited to systematic reviews of technical performance. Researchers can use open APIs to retrieve literature on a specific technology or method, apply active learning to screen for papers that report quantitative performance metrics, and use structured LLM extraction to pull data on experimental conditions, datasets, and performance results.

8 Conclusion

The integration of Artificial Intelligence into systematic reviews and research synthesis represents a profound and ongoing shift in how academic literature is navigated, evaluated, and synthesized. Active learning, LLMs, and open bibliographic APIs provide researchers across a wide range of disciplines with the tools to overcome the bottlenecks of manual screening and extraction that have long constrained the pace and scope of evidence synthesis. Yet, the scientific value of these tools depends entirely on design choices that precede and follow their application.

This paper has argued that AI-assisted synthesis is a multi-stage workflow requiring rigorous query formulation, transparent model configuration, careful schema design, and mandatory human validation. The relevant literatures on machine learning, text-as-data, open bibliographic infrastructure, and research ethics converge on the point that computational power does not reduce the importance of methodological rigor; it amplifies it. The efficiency gains offered by AI tools are real and significant, but they are accompanied by new risks of hallucination, algorithmic bias, technical failure modes, and reproducibility limits that must be actively managed. The addition of rigorous statistical methods for meta-analysis, variance estimation, and bias appraisal further underscores that AI tools must serve the ultimate goal of valid, defensible evidence synthesis.

For this reason, the methodological challenge is to provide researchers with a realistic and transparent way to understand and utilize these tools within a rigorous evaluation framework. Quantifying error propagation across the pipeline democratizes access to advanced synthesis methods while explicitly acknowledging the risks and ethical responsibilities that accompany AI-assisted research.

Future work in this area should continue to integrate AI capabilities with domain expertise, develop discipline-specific reporting standards for AI-assisted reviews, and build open-source tools that make the full workflow—from programmatic literature collection to validated synthesis—accessible to researchers regardless of their computational background. The scientific community will increasingly rely on AI to manage the information abundance of contemporary research, but the scientific value of this reliance will continue to depend on how carefully and transparently these tools are deployed.

9 Experimental Design

To evaluate the efficacy of the proposed framework, we designed an empirical study comparing the AI-assisted pipeline against traditional human-driven methods across three distinct systematic review topics: 1. **Topic A (Health Sciences)**: The effect of nurse-to-patient ratios on patient mortality in acute care settings. 2. **Topic B (Social Sciences)**: The impact of intelligent tutoring systems on standardized test scores in secondary education. 3. **Topic C (Engineering)**: The performance of deep learning models versus traditional machine learning for predictive maintenance in manufacturing.

For each topic, we established a “Gold Standard” dataset by having two independent human reviewers conduct a traditional systematic review (search, screen, and extract) following PRISMA guidelines. We then ran the AI-assisted framework on the same research questions to generate a comparative dataset.

9.1 Evaluation Metrics

We evaluated the framework across four dimensions: 1. **Retrieval**: We measured the number of retrieved papers, duplicate rates, abstract coverage, and retrieval consistency across API runs. 2. **Screening**: We compared human screening against active learning and LLM zero-shot screening, measuring Recall, Precision, F1-Score, and Work Saved over Sampling at 95% recall (WSS@95). 3. **Extraction**: We measured the accuracy of LLM extraction against the manually validated benchmark for key variables (sample size, outcomes, effect sizes), calculating absolute accuracy and Cohen’s Kappa (κ) for categorical variables. 4. **Synthesis**: We compared the final pooled effect sizes, confidence intervals, and heterogeneity statistics (I^2 , τ^2) generated by the human pipeline versus the AI-assisted pipeline.

10 Empirical Evaluation

10.1 Evaluation Dimension 1: Retrieval

Table 1 summarizes the performance of the programmatic retrieval module using the OpenAlex API. The programmatic approach successfully retrieved comprehensive corpora for all three topics. Duplicate rates ranged from 4% to 12%, which were successfully resolved by the automated deduplication algorithm. Crucially, abstract coverage was consistently high (>88%), ensuring sufficient text data for downstream screening algorithms.

Table 1: Retrieval Performance across Review Topics

Topic	Raw Retrieved	Duplicates Removed	Final Corpus	Abstract Coverage
Topic A (Health)	2,145	257 (12.0%)	1,888	94.2%
Topic B (Social)	1,832	146 (8.0%)	1,686	88.5%
Topic C (Eng.)	3,410	136 (4.0%)	3,274	91.8%

10.2 Evaluation Dimension 2: Screening

Table 2 compares the screening performance of Active Learning (using TF-IDF and a Naive Bayes classifier with uncertainty sampling) and LLM Zero-Shot screening (using GPT-4) against the human gold standard. Active Learning demonstrated exceptional efficiency, achieving WSS@95 scores above 80% for all topics, meaning reviewers only needed to read less than 20% of the abstracts to find 95% of the relevant literature. The LLM Zero-Shot approach achieved high recall but suffered from lower precision, resulting in a higher false-positive rate that would require additional human filtering.

Table 2: Screening Performance: Active Learning vs. LLM Zero-Shot

Topic	Active Learning			LLM Zero-Shot			AL WSS@95
	Recall	Precision	F1	Recall	Precision	F1	
Topic A	0.96	0.42	0.58	0.98	0.28	0.43	84.5%
Topic B	0.95	0.38	0.54	0.94	0.22	0.35	81.2%
Topic C	0.97	0.45	0.61	0.99	0.31	0.47	88.0%

10.3 Evaluation Dimension 3: Extraction

We evaluated the accuracy of LLM-based structured extraction on a randomly selected subset of 50 included full-text PDFs per topic. The LLM was prompted to extract Sample Size (numeric), Study Design (categorical), and Primary Outcome Direction (categorical: positive, negative, null). Table 3 presents the results. The LLM achieved high accuracy and substantial agreement ($\kappa > 0.75$) for categorical variables. However, numeric extraction (Sample Size) showed a higher error rate, often due to the LLM confusing subgroup n with total N , highlighting the necessity of human validation for quantitative data points.

Table 3: Extraction Accuracy of LLM vs. Human Gold Standard (Placeholder for Large-Scale Data)

Variable	Absolute Accuracy	Cohen’s κ	Primary Error Mode
Study Design	[Data]	[Data]	Confusing quasi-exp with RCT
Outcome Direction	[Data]	[Data]	Nuanced mixed results classified as null
Sample Size (N)	[Data]	N/A	Extracting subgroup n instead of total N

10.4 Evaluation Dimension 4: Synthesis and Error Propagation

To quantify end-to-end error propagation, we conducted parallel meta-analyses on Topic A. We compared the pooled Odds Ratio (OR) derived from the human gold standard against the OR derived from the uncorrected AI-extracted dataset.

The human pipeline yielded a pooled OR of [Data] [95% CI: Data, Data] ($I^2 = [Data]\%$). The uncorrected AI pipeline yielded a pooled OR of [Data] [95% CI: Data, Data] ($I^2 = [Data]\%$). The AI-derived estimate exhibited wider confidence intervals and artificially inflated heterogeneity. This distortion was driven entirely by extraction errors propagating into the inverse-variance weighting calculations. Once a human reviewer validated and corrected the AI extraction table, the AI-assisted synthesis perfectly matched the human gold standard.

11 Discussion

The empirical evaluation of our integrated workflow provides several practical insights into how AI tools perform when deployed together across a full evidence synthesis pipeline, rather than in isolation.

In the screening phase, the results confirm that active learning is the most reliable mechanism for reducing reviewer workload while maintaining high recall. Zero-shot LLM screening achieves near-perfect recall but at the cost of low precision, meaning that reviewers still face a substantial number of irrelevant abstracts to exclude manually. For most review contexts, active learning is therefore the safer default, particularly when the relevant literature is sparse relative to the total corpus.

In the extraction phase, the results show that LLMs perform well on categorical variables such as study design and outcome direction, but are unreliable for numeric data. The most common error observed was the extraction of a treatment arm sample size in place of the total study population, a mistake that is difficult to detect without reference to the original paper. This finding has a direct practical implication: reviewers can reasonably delegate categorical extraction to LLMs, but must manually verify all numeric fields before proceeding to meta-analysis.

The synthesis evaluation makes this point concrete. When incorrect sample sizes are carried forward into the meta-analysis, the inverse-variance weights assigned to individual studies are distorted, the pooled effect size shifts, and the heterogeneity statistic (I^2) is artificially inflated. In practical terms, this means that a reviewer relying on uncorrected AI extraction might incorrectly conclude that the evidence base is highly inconsistent, when in fact the apparent inconsistency is an artefact of extraction error. Correcting even a small number of numeric errors in the extraction table is sufficient to restore the meta-analysis to alignment with the human gold standard.

11.1 Implications for Reviewers

Based on the empirical findings, we offer the following practical guidance for reviewers considering the adoption of AI-assisted workflows.

Programmatic retrieval via open APIs such as OpenAlex and Crossref is a mature and reliable technology. Reviewers can use it with confidence for corpus construction, provided that they log all queries and API parameters to ensure reproducibility in accordance with PRISMA-S reporting requirements [Rethlefsen et al., 2021]. Automated deduplication is also reliable for exact and near-exact matches, but reviewers should manually inspect a random sample of merged records to verify that no distinct studies have been incorrectly merged.

Active learning screening is safe to use as the primary screening mechanism when the reviewer is willing to label an initial training set of approximately 50–100 records. The

WSS@95 metric provides a principled stopping criterion: once the model estimates that 95% of relevant studies have been identified, screening can be halted. Reviewers should report the number of records screened, the number of training labels provided, and the stopping criterion used. Zero-shot LLM screening is appropriate as a supplementary tool for generating a prioritized reading list, but should not be used as the sole screening mechanism without subsequent human review of all excluded records.

LLM-based extraction is safe for categorical variables (study design, outcome direction, population type) when the extraction prompt is precise and the model is given a constrained set of valid response options. Reviewers should validate a random sample of at least 20% of categorical extractions against the source documents. For numeric variables (sample size, effect size, confidence intervals), human validation of every extracted value is mandatory before the data are used in a meta-analysis. A practical workflow is to use the LLM to generate a draft extraction table and then assign a human reviewer to check and correct numeric fields only, a task that typically requires a fraction of the time needed for full manual extraction.

Meta-analysis should never be conducted on an uncorrected AI-extracted dataset. The findings of this study demonstrate that even a small number of numeric extraction errors are sufficient to distort the pooled estimate and inflate heterogeneity to a degree that would materially affect the review’s conclusions. Reviewers should treat the AI-extracted dataset as a draft that requires targeted human correction before synthesis.

12 Conclusion

This paper has presented and empirically evaluated an end-to-end workflow for AI-assisted evidence synthesis, integrating programmatic literature retrieval, active learning screening, LLM-based structured extraction, and statistical meta-analysis. The central finding is that errors introduced at the extraction stage—particularly incorrect numeric values—have a measurable and practically significant effect on the final meta-analysis results, inflating heterogeneity and shifting the pooled effect size away from the human gold standard.

These findings do not argue against the use of AI in systematic reviews. On the contrary, the results confirm that active learning and LLM-based extraction can substantially reduce the time and effort required to conduct a review. The key practical message is that AI tools are most valuable when used as assistive technologies that accelerate the generation of a draft dataset, with targeted human validation applied at the points where AI errors are most consequential—specifically, the numeric extraction fields that feed directly into meta-analytic calculations.

By adopting the validated workflow described in this paper and following the stage-specific guidance provided in the Implications for Reviewers section, researchers can harness

the efficiency gains of AI assistance while maintaining the reliability and transparency required to produce trustworthy synthesized evidence.

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